

Flyback and Snubber RC The Practical Guide to Safe Inductive Circuits



Leonardo Chieco

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The practical guide to safe inductive circuits

In this tutorial, we will explore how to protect mosfets and other switching devices from voltage spikes and oscillations generated by inductive loads. We will analyze the operation of flyback diodes, Zener diodes, and RC snubbers, showing how each of these elements affects turn-off speed, energy dissipation, and device protection. We will understand why inductive transients can cause damage, how oscillations are generated, and which strategies can be adopted to limit them, with practical examples and real-world references to circuits using micro-valves or fast relays.

During the analysis, simulations were performed using LTspice in order to directly observe voltage waveforms, current behavior, and ringing phenomena. Readers can easily reproduce these schematics and experiment independently by modifying circuit parameters, gaining a deeper understanding of how different protection techniques influence system behavior.

This approach will enable you to design safer, more reliable, and faster systems, with both theoretical and practical insight into the phenomena involved.

Enjoy the reading!

Author

Leonardo Chieco is an electronic engineer with over 20 years of experience in automation control software development (PC/PLC), electronic board design for industrial applications, firmware development, robotics, and mechatronics.

LinkedIn: <https://www.linkedin.com/in/leonardo-chieco-53550b129/>

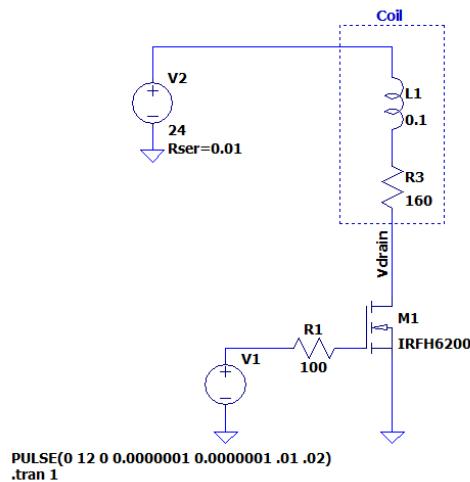
1. Overvoltage Suppression Techniques

When driving a DC coil, the main issue during turn-off is managing the energy stored in the inductance. This energy, equal to:

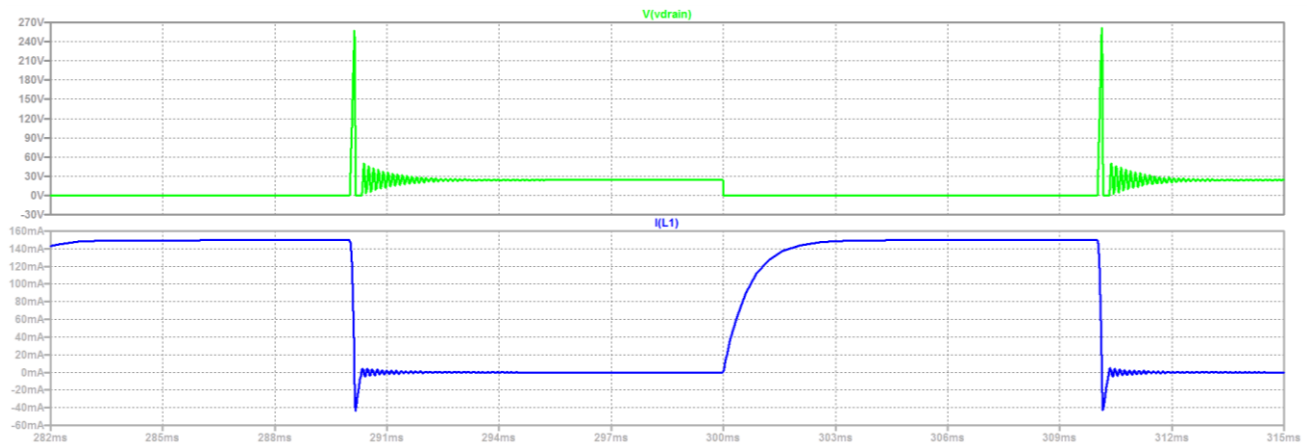
$$E_L = \frac{1}{2} L I_0^2$$

cannot disappear instantaneously and manifests itself as an overvoltage across the switching device (whether mechanical or electronic). Depending on the protection circuit used, the dynamic behavior changes dramatically.

Let us assume the coil is driven by a power mosfet as shown in the following figure.



The LTspice simulation immediately shows the effects of E_L on both the current flowing through the coil and the voltage at the mosfet drain. We can observe that the drain voltage rises up to approximately 250 V, which could lead to device failure. In addition, oscillations appear that may generate electromagnetic interference (EMI).



At the switching-off instant, the inductance contains an energy $E_L = \frac{1}{2} L I_0^2$ which cannot simply vanish. So where is this energy dissipated?

1. In the mosfet (most common case).

The drain voltage rises until the mosfet enters avalanche breakdown. In this condition, the mosfet conducts at high voltage; current continues flowing for a short time, and the energy E_L is dissipated in the silicon. This is one of the reasons why many mosfets “die for no apparent reason.

2. In parasitic capacitances.

The drain node always has an output capacitance C_{OSS} , and PCB traces introduce parasitic capacitances to ground that are difficult to quantify. Part of the energy is therefore transferred into capacitive form $E_C = \frac{1}{2} C V^2$ causing voltage overshoot, LC ringing, and electromagnetic emissions. Ultimately, this energy is still dissipated in parasitic resistances and dielectric losses.

3. In arcs or discharges (worst-case scenario).

If the voltage becomes high enough, sparks and air ionization may occur. This happens in relays and contactors switching inductive loads.

Why does the current drop so quickly? The current decreases very rapidly because the voltage across the inductor can rise to very high values. The fundamental inductor relationship always applies:

$$\frac{di(t)}{dt} = - \frac{V_L}{L}$$

When the mosfet turns off and no controlled current path exists, the inductor generates whatever voltage is necessary to oppose the change in current. If the voltage V_L becomes very large (in absolute value), then the term $\frac{di(t)}{dt}$ also becomes large, forcing the current to decrease very rapidly.

In other words, the higher the voltage generated by the inductor, the faster the current variation. Without a clamping element, the voltage may rise until it reaches the limits imposed by circuit parasitics or the mosfet breakdown voltage, resulting in an extremely fast but potentially destructive turn-off.

For this reason, several techniques exist to limit the generated overvoltage and to safely control the transient behavior. Let us now examine them in detail.

Flyback diode

A first and very common solution to this problem is the use of a flyback diode, connected between the drain and the supply rail, in parallel with the coil. When the mosfet turns off and the voltage tends to rise, the diode immediately enters conduction and clamps the overvoltage to:

$$V_{DRAIN} = V_{BUS} + V_{F,DIODE}$$

The coil current continues to circulate in a closed loop formed by the coil, the diode, and the power supply. In this configuration, the energy is mainly dissipated in the coil resistance and in the diode's forward voltage drop.

The advantage of this solution is very effective mosfet protection and reduced electromagnetic emissions. The drawback is that the discharge voltage is very low (approximately 0.7 V), so the current decreases slowly according to:

$$L \frac{di(t)}{dt} + R_{COIL}i(t) + V_{F,DIODE} = 0$$

$$\frac{di(t)}{dt} + \frac{R_{COIL}}{L}i(t) = - \frac{V_{F,DIODE}}{L}$$

The general solution to this equation is:

$$i(t) = A e^{-\frac{R}{L}t} + B$$

Let us determine A and B. We start from the relation:

$$\frac{di(t)}{dt} + \frac{R_{COIL}}{L} i(t) = - \frac{V_{F,DIODE}}{L}$$

Now we substitute $i(t)$:

$$-\frac{R_{COIL}}{L} A e^{-\frac{R_{COIL}}{L}t} + \frac{R_{COIL}}{L} (A e^{-\frac{R_{COIL}}{L}t} + B) = - \frac{V_{F,DIODE}}{L}$$

$$\frac{R_{COIL}}{L} B = - \frac{V_{F,DIODE}}{L}$$

$$B = - \frac{V_{F,DIODE}}{R_{COIL}}$$

Now we impose the condition $i(t) = I_0$ which is the current flowing in the coil just before the mosfet turns off.

$$i(0) = A e^{-\frac{R}{L}0} - \frac{V_{F,DIODE}}{R_{COIL}} = I_0$$

$$A - \frac{V_{F,DIODE}}{R_{COIL}} = I_0$$

$$A = \frac{V_{F,DIODE}}{R_{COIL}} + I_0$$

Therefore:

$$i(t) = \left(\frac{V_{F,DIODE}}{R_{COIL}} + I_0 \right) e^{-\frac{R_{COIL}}{L}t} - \frac{V_{F,DIODE}}{R_{COIL}}$$

This equation is valid only as long as the assumptions under which it was derived remain satisfied, namely, as long as the diode is conducting and can be modeled with a constant forward voltage drop $V_{F,DIODE}$.

When the current decreases to zero, the diode becomes reverse-biased and turns off. From that instant onward, the circuit configuration changes and the previous differential equation is no longer applicable.

The formal mathematical solution suggests that:

$$i(t \rightarrow \infty) = - \frac{V_{F,DIODE}}{R_{COIL}}$$

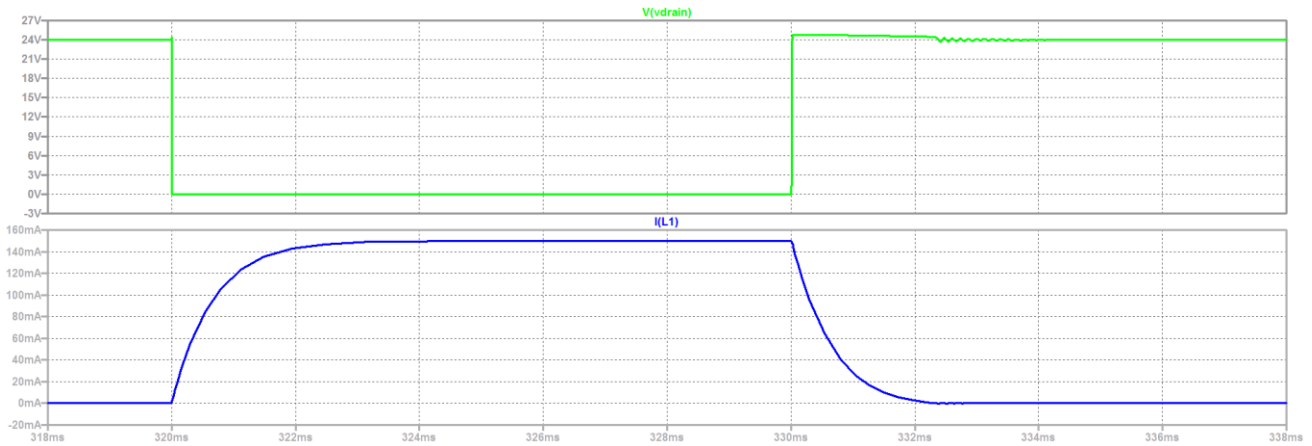
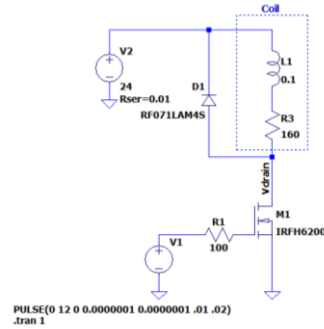
However, this result is purely theoretical and arises from extending the solution beyond its domain of validity.

Physically, a negative current would imply a reversal of current flow through the diode, which is impossible because the device is turned off under those conditions.

Therefore, the real current never becomes negative: it gradually decreases to zero and remains zero for times greater than the diode turn-off instant. By imposing $i(t) = 0$ we obtain:

$$t_{zero} = \frac{L}{R_{COIL}} \ln \left(\frac{\frac{V_{F,DIODE}}{R_{COIL}} + I_0}{\frac{V_{F,DIODE}}{R_{COIL}}} \right)$$

Let us simulate this case in LTspice.



As we can see, the drain voltage of the mosfet no longer exhibits dangerous spikes, but the current in the coil takes more than two milliseconds to extinguish. For a typical relay, this delay is negligible. However, for a micro-valve driven in PWM mode, it could represent a significant issue.

Why does this happen?

Recall that:

$$\frac{di(t)}{dt} = - \frac{V_L}{L}$$

In this case, the inductor voltage V_L is approximately equal to the diode forward voltage $V_{F,DIODE} \simeq 0.7V$. Therefore, $i(t)$ decreases much more slowly compared to the “no-diode” case, where V_L could reach very high values.

Flyback diode and zener diode

We have seen that when using a simple flyback diode, the voltage across the coil during turn-off is clamped to $V_{F,DIODE}$. This perfectly protects the mosfet, but results in a very slow discharge.

To speed up turn-off without sacrificing protection, a Zener diode can be placed in series with the flyback diode (or alternatively, a unidirectional TVS can be used). In this configuration, when the mosfet turns off, the voltage is no longer clamped to 0.7 V but to:

$$V_L = V_{F,DIODE} + V_{ZENER}$$

For example, if the Zener is rated at 33 V, the coil will discharge at approximately 34 V.

This completely changes the dynamics: the voltage across the inductor is much higher compared to the simple diode case, therefore the current decreases much more rapidly. The valve closes sooner, the relay releases faster, and the de-energization time is drastically reduced.

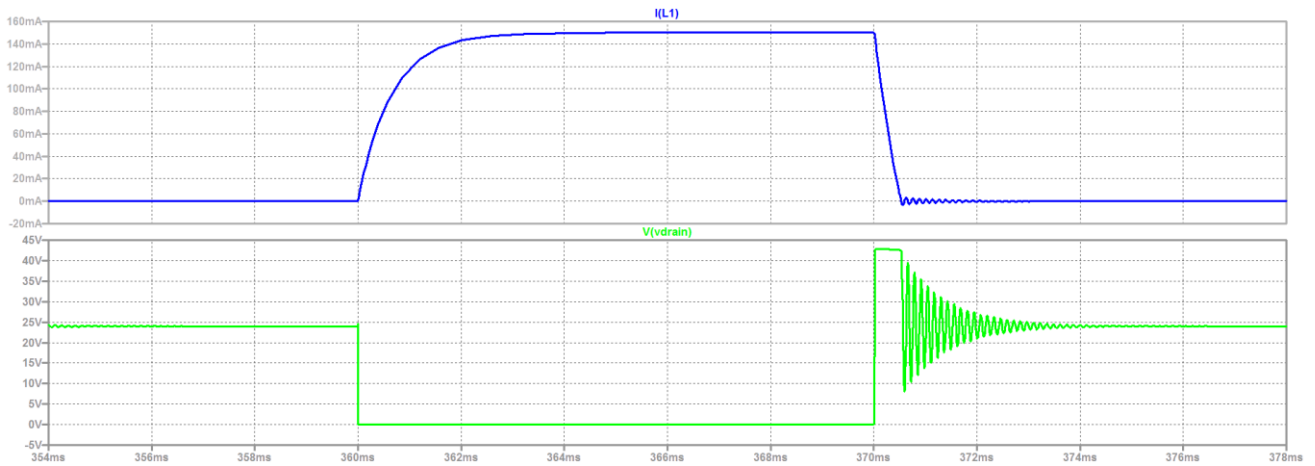
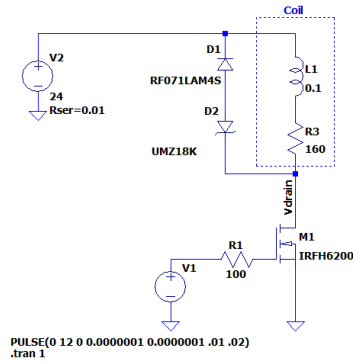
From an energy standpoint, E_L is mainly dissipated in the Zener diode, which must therefore be sized to withstand the pulsed energy. The mosfet remains protected because the voltage is limited to a controlled value selected by the designer.

This configuration is used when reducing the release time of the inductive load is important, for example:

- Fast pneumatic micro-valves
- Injectors
- Relays in fast control systems
- Actuators where closing time affects system precision

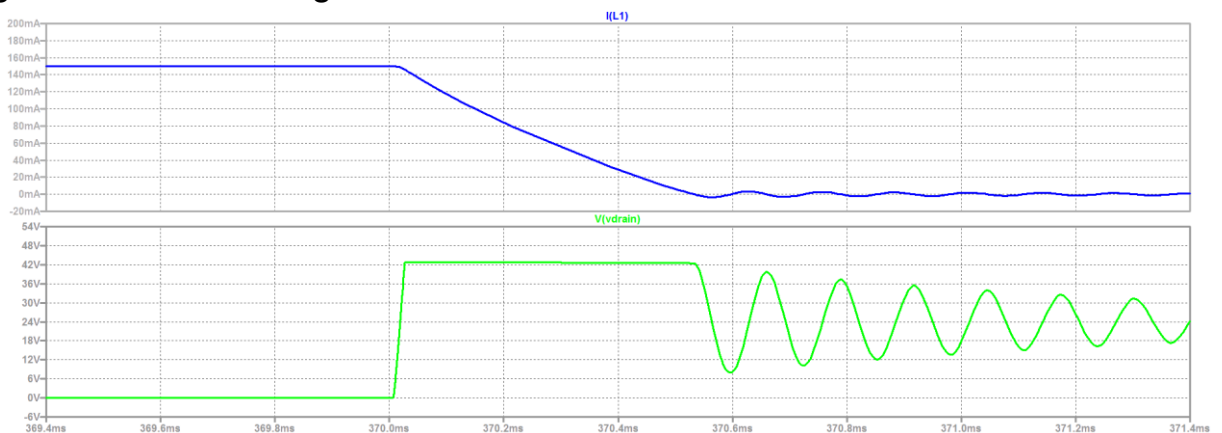
In practice, it represents an elegant compromise between mosfet safety and dynamic system performance.

Let us simulate this condition.



We can observe that:

1. The discharge process is significantly faster.
2. The maximum voltage increases in a controlled manner according to the previously derived relation.
3. Oscillations appear as soon as the current reaches zero, which may generate electromagnetic interference.

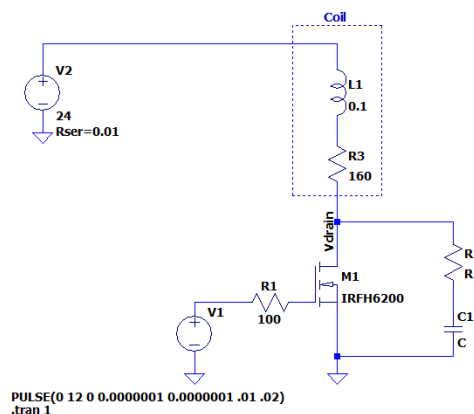


When the mosfet is off and the coil current falls below the level required to keep the zener (or flyback diode) conducting, the controlled discharge path is interrupted. However, the residual energy stored in the coil cannot simply disappear. An inductor stores energy in the form of current, so if no conductive path is available, the energy must find alternative ways to be converted.

What happens is that the remaining energy is transferred to the circuit's parasitic capacitances (primarily the mosfet's C_{oss}) and dissipated through the resistive losses present in the system. This forms a damped LC circuit that produces voltage and current oscillations until the residual energy is fully dissipated as heat.

The RC Snubber

The RC snubber is a widely used circuit in power electronics to protect switching devices such as mosfet or IGBTs from voltage stress caused by transients. It consists of a resistor and a capacitor connected in parallel with the mosfet, that is, between drain and source (or collector and emitter). It has two main functions: damping high-frequency oscillations caused by the interaction between the circuit's parasitic inductance and the mosfet output capacitance (C_{oss}), and limiting the voltage spike across the device during turn-off.



The energy stored in the inductance is $E_L = \frac{1}{2} L I_0^2$ where I_0 is the current at the turn-off instant.

When the mosfet (or switch) opens, this energy must go somewhere, namely into the snubber.

The role of the capacitor C is to absorb the inductor's energy and limit the maximum voltage V_{MAX} at the mosfet drain (i.e., between the switching node and ground).

The energy stored in the capacitor will be:

$$E_C = \frac{1}{2} C V_{MAX}^2$$

Design condition:

$$E_L \leq E_C$$

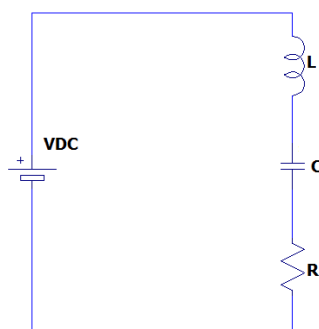
$$\frac{1}{2} L I_0^2 \leq \frac{1}{2} C V_{MAX}^2$$

so:

$$C \geq \frac{L I_0^2}{V_{MAX}^2}$$

This is the most important formula.

When the mosfet turns off, a series RLC circuit is formed.



Applying Kirchhoff's voltage law:

$$V_{DC} = v_L + v_C + v_R$$

Therefore

$$V_{DC} = L \frac{di}{dt} + \frac{1}{C} \int i dt + R i$$

Differentiating with respect to time to eliminate the integral:

$$\frac{d V_{DC}}{dt} = L \frac{d^2 i}{dt^2} + \frac{1}{C} i + R \frac{d i}{dt}$$

but

$$\frac{d V_{DC}}{dt} = 0$$

Because V_{DC} is constant.

Therefore:

$$L \frac{d^2 i}{dt^2} + \frac{1}{C} i + R \frac{d i}{dt} = 0$$

Dividing by L :

$$\frac{d^2 i}{dt^2} + \frac{R}{L} \frac{d i}{dt} + \frac{1}{CL} i = 0$$

Solving the Differential Equation, we look for solutions of the form:

$$i(t) = e^{st}$$

Substituting:

$$\frac{d^2 i}{dt^2} + \frac{R}{L} \frac{d i}{dt} + \frac{1}{CL} i = s^2 e^{st} + \frac{R}{L} s e^{st} + \frac{1}{CL} e^{st} = 0$$

Dividing by e^{st} (always different from zero):

$$s^2 + \frac{R}{L} s + \frac{1}{CL} = 0$$

Solving the quadratic equation to find the solutions s_1 and s_2 .

$$s_{1,2} = -\frac{R}{2L} \pm \sqrt{\left(\frac{R}{2L}\right)^2 - \frac{1}{CL}} = -\frac{R}{2L} \pm \frac{1}{\sqrt{LC}} \sqrt{\frac{R^2 C}{4L} - 1}$$

We define:

- Natural angular frequency: $\omega_0 = \frac{1}{\sqrt{LC}}$
- Damping factor: $\zeta = \frac{R}{2} \sqrt{\frac{C}{L}}$

Then:

$$s_{1,2} = -\zeta\omega_0 \pm \omega_0 \sqrt{\zeta^2 - 1}$$

There are three possible conditions.

Underdamped System ($\zeta < 1$)

The roots are complex.

$$s_{1,2} = -\zeta\omega_0 \pm j \omega_0 \sqrt{1 - \zeta^2}$$

It would seem, at first glance, that there are two distinct solutions:

$$i_1(t) = e^{s_1 t}, \quad i_2(t) = e^{s_2 t}$$

But this is just a special case. Any combination of these solutions is also a solution:

$$i(t) = A e^{s_1 t} + B e^{s_2 t}$$

Let's see in detail what $i(t)$ is worth:

$$i(t) = Ae^{s_1 t} + Be^{s_2 t} = Ae^{(-\zeta\omega_0 + j\omega_0\sqrt{1-\zeta^2})t} + Be^{(-\zeta\omega_0 - j\omega_0\sqrt{1-\zeta^2})t}$$

$$i(t) = e^{-\zeta\omega_0 t} (Ae^{(j\omega_0\sqrt{1-\zeta^2})t} + Be^{(-j\omega_0\sqrt{1-\zeta^2})t})$$

Using Euler's formulas ($e^{\pm ja} = \cos(a) \pm j \sin(a)$), we obtain:

$$i(t) = e^{-\zeta\omega_0 t} (A \cos \omega_D t + j A \sin \omega_D t + B \cos \omega_D t - j B \sin \omega_D t)$$

$$i(t) = e^{-\zeta\omega_0 t} (P_1 \cos \omega_D t + j P_2 \sin \omega_D t)$$

The current exhibits a damped oscillation, but this is precisely what we want to avoid. Therefore, the condition $\zeta < 1$ must be avoided.

Critically Damped System ($\zeta=1$)

$$s_{1,2} = -\zeta\omega_0$$

So:

$$i(t) = e^{-\zeta\omega_0 t}$$

As in the previous case, it can be shown that if $e^{-\zeta\omega_0 t}$ is a solution, then $t e^{-\zeta\omega_0 t}$ will also be a solution and therefore, even a combination of them is still a solution. Ultimately:

$$i(t) = P_1 e^{-\zeta\omega_0 t} + P_2 t e^{-\zeta\omega_0 t} = e^{-\zeta\omega_0 t} (P_1 + P_2 t)$$

Note that, in this case, the current approaches zero without oscillations.

The dominant term defining the response speed is $e^{-\zeta\omega_0 t}$. Therefore, the time constant is:

$$\tau = \frac{1}{\zeta\omega_0}$$

Overdamped System ($\zeta > 1$)

In this case we have two real solutions.

$$s_{1,2} = -\zeta\omega_0 \pm \omega_0 \sqrt{\zeta^2 - 1}$$

The general solution of the differential equation is:

$$i(t) = Ae^{s_1 t} + Be^{s_2 t} = A e^{-\zeta\omega_0 + \omega_0 \sqrt{\zeta^2 - 1} t} + B e^{-\zeta\omega_0 - \omega_0 \sqrt{\zeta^2 - 1} t}$$

$$i(t) = A e^{-(\zeta\omega_0 - \omega_0 \sqrt{\zeta^2 - 1}) t} + B e^{-(\zeta\omega_0 + \omega_0 \sqrt{\zeta^2 - 1}) t}$$

Where s_1 and s_2 are certainly negative, so $i(t)$ decreases towards zero. The convergence rate is certainly slower than in the case of coincident roots.

Let's see why.

The term with B decreases very quickly because:

$$\zeta\omega_0 - \omega_0 \sqrt{\zeta^2 - 1} \ll \zeta\omega_0 + \omega_0 \sqrt{\zeta^2 - 1}$$

Therefore, it is the term with A that sets the dynamics with a time constant:

$$\tau = \frac{1}{\zeta\omega_0 - \omega_0 \sqrt{\zeta^2 - 1}}$$

Which is certainly higher than that of the previous case ($\frac{1}{\zeta\omega_0}$).

The system, therefore, does not oscillate, but tends to respond more slowly.

From this quick analysis, it is clear that it is best to work close to $\zeta = 1$.

Practical example

For a snubber, the goal is to absorb all the inductor energy in the shortest possible time without oscillations. Mathematically, this means:

- decreasing monotonic response
- minimum settling time
- no ringing (spurious oscillations)

This only applies to critical damping.

$$\zeta = 1 \rightarrow R = 2 \sqrt{\frac{L}{C}}$$

Known data:

- Inductance seen at shutdown: L
- Shutdown current: $I_0 = 150 \text{ mA}$
- Maximum voltage on the switch (mosfet): $V_{DS,MAX} = 60V$
- Supply voltage: $V_{DC} = 24V$

$$V_{snub} = V_{Dmax} - V_{DC}$$

$$V_{snub} = 60 - 24 = 36 \text{ V}$$

The energy of the inductor must all go into the capacitor:

$$\frac{1}{2} L I^2 = \frac{1}{2} C V_{snub}^2$$

$$C = \frac{L I^2}{V_{snub}^2} = \frac{0.05 \cdot 0.15^2}{36^2} = 1 \mu F \quad (V_{rated} > 1.5 \dots 2 V_{snub})$$

Now let's calculate the resistance

$$R_{critic} = 2 \sqrt{\frac{L}{C}} = 2 \sqrt{\frac{0.05 \text{ H}}{1 \mu F}} = 500 \Omega$$

The peak current in the resistor is approximately:

$$I_{PK} = \frac{V_{MAX}}{R}$$

The average power that R must dissipate is equal to that stored by the inductance averaged over time:

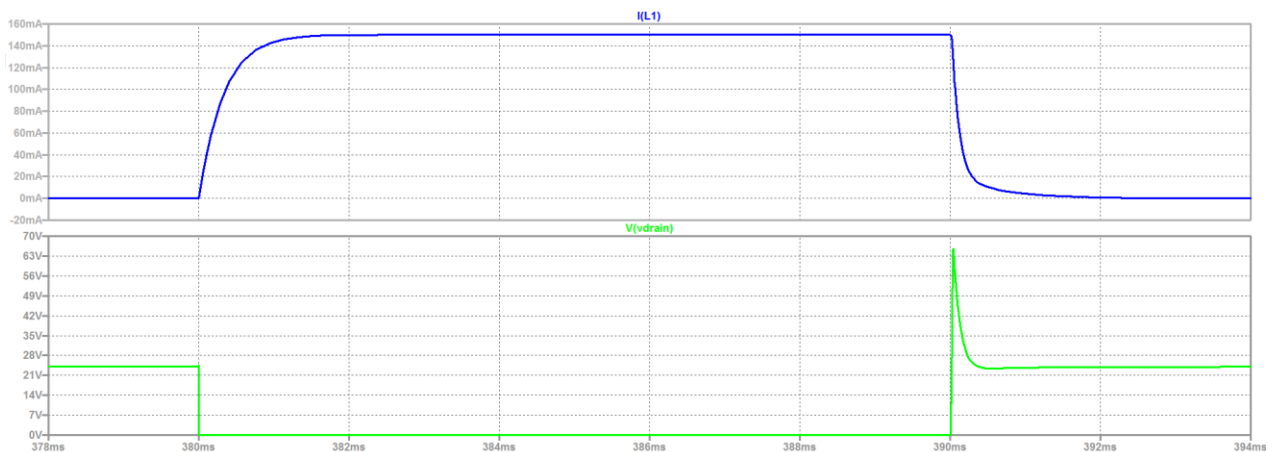
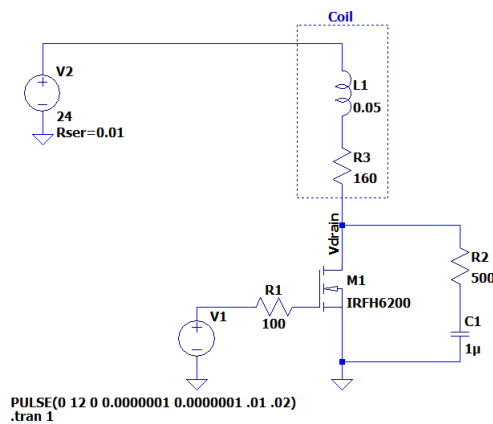
$$P_R = \frac{1}{2} L I_0^2 f_s$$

Where f_s is the switching frequency.

$$P_R = E_L \cdot f = \frac{1}{2} L I^2 \cdot f = 0.0006 \cdot 50 \text{ Hz} = 0.03 \text{ W}$$

A quarter-watt resistor might work well.

Let's simulate with LTSpice.



You can see that the response is fast and the drain voltage is controlled.

RC snubber in parallel with the coil

When the RC snubber is placed in parallel with the coil, its purpose is not so much to directly protect the mosfet from high-frequency ringing (as in the case of the drain snubber), but to provide a controlled path for the inductor energy during turn-off.

Consider a coil of inductance L carrying a current I_0 when the mosfet turns off. The stored energy is:

$$E_L = \frac{1}{2} L I_0^2$$

When the switch opens, the current cannot instantly disappear, so it closes on the branch consisting of R and C in series, connected in parallel to the coil.

By applying Kirchhoff's law:

$$v_L = v_R + v_C$$

with:

$$v_L = L \frac{di(t)}{dt} = R i(t) + \frac{1}{C} \int i(t) dt$$

By differentiating to eliminate the integral, we obtain the second order differential equation:

$$L \frac{d^2 i(t)}{dt^2} = R \frac{d i(t)}{dt} + \frac{1}{C} i(t)$$

This is the classic equation for a series-damped RLC system and is solved similarly to the previous one.

The fundamental parameters are:

- Natural pulsation: $\omega_0 = \frac{1}{\sqrt{LC}}$
- Damping factor: $\zeta = \frac{R}{2} \sqrt{\frac{C}{L}}$

The snubber's behavior depends on ζ :

- $\zeta < 1 \rightarrow$ damped oscillations (to be avoided)
- $\zeta = 1 \rightarrow$ critical damping (no oscillations)
- $\zeta > 1 \rightarrow$ overdamped (no oscillations, but slow response)

Physically, when the circuit is turned off:

1. The energy of the inductor charges the capacitor.
2. The capacitor discharges into the resistor.
3. The energy is dissipated as heat in R.

The voltage on the coil (and therefore on the mosfet) does not rise freely but is controlled by the capacitor:

$$v_c(t) = \frac{1}{C} \int i(t) dt$$

The maximum voltage value can be estimated by imposing that all the magnetic energy is transferred to the capacitor (ideal case):

$$E_L = E_C$$

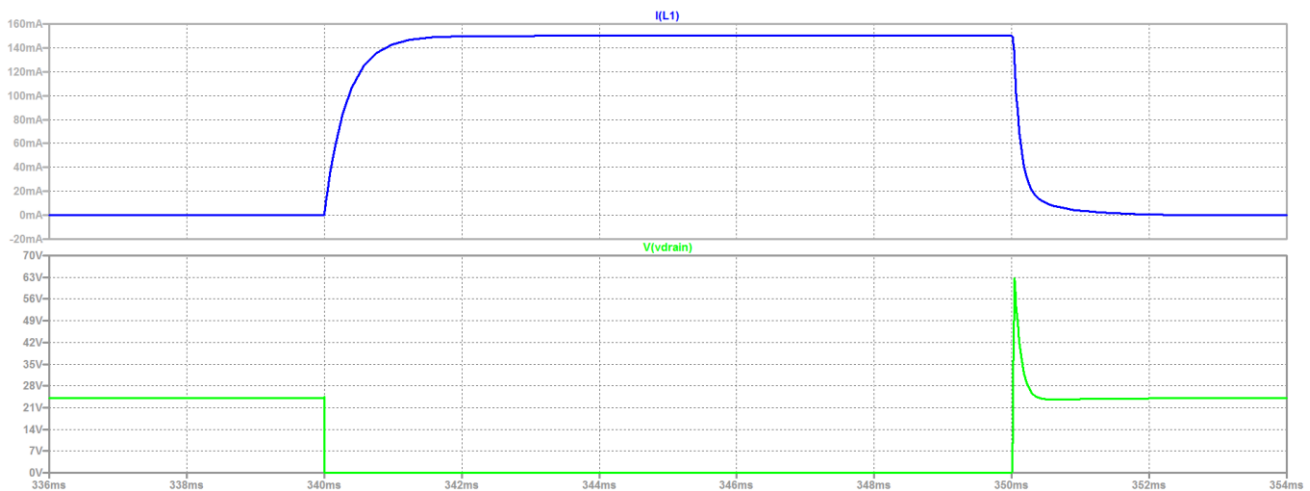
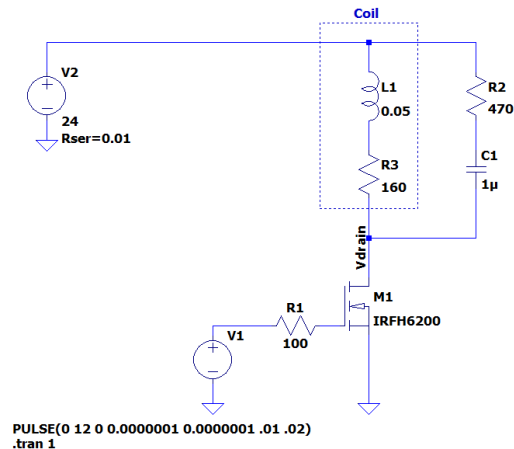
$$\frac{1}{2} L I_0^2 = \frac{1}{2} C V_{MAX}^2$$

from which:

$$V_{MAX} = \sqrt{\frac{L I_0^2}{C}} = I_0 \sqrt{\frac{L}{C}}$$

This relationship is fundamental for capacitor sizing. The snubber coil is used:

- When faster turn-off is required than with flyback
- When voltage is to be limited without using a Zener
- When energy is to be controlled directly at the load
- In circuits where EMI must be controlled locally



Type of capacitor to use for the snubber

In the RC snubber, the capacitor is not a simple filter element, but a component that must absorb very rapid and repetitive energy pulses, with high $\frac{dV}{dt}$ values and significant pulse currents. For this reason, the choice of technology is crucial. In power electronics, the most suitable solution is almost always the polypropylene film capacitor (MKP), because it offers low ESR, low parasitic inductance (ESL), excellent long-term stability, and excellent ability to withstand high voltage

gradients. Polypropylene also has very low dielectric losses, so it dissipates less internal power when absorbing the energy coming from the inductance when the device is turned off. Polyester film capacitors (MKT) can be used in less demanding applications, but have higher losses and less favorable behavior at high frequencies. Electrolytic capacitors, on the other hand, are not suitable as direct snubbers on mosfet or IGBTs due to their high ESR and limited pulse current carrying capacity.

Even multilayer ceramic capacitors (MLCC) can only be used for small capacitance values and to dampen very high-frequency ringing, but keep in mind the significant reduction in capacitance with applied DC voltage. In any case, it is essential to choose an adequate nominal voltage (typically at least 1.5–2 times the maximum expected voltage) and to carefully design the snubber, keeping connections very short to minimize parasitic inductances that would compromise the snubber's effectiveness.

2. Conclusions

When driving an inductive load with a mosfet, the real problem is not turn-on, but turn-off.

The energy stored in the inductor cannot disappear instantly; it must be converted or dissipated somewhere.

Depending on how we manage this energy, we obtain very different behaviors:

- If we don't provide a controlled path, the energy generates a high voltage on the mosfet.
- If we use a flyback diode, the energy slowly dissipates in the coil's resistance.
- If we use a Zener or clamp diode, we accelerate turn-off by increasing the controlled voltage.
- If we use an RC snubber, we dampen oscillations and ringing due to parasitic oscillations.

There is no "always best" solution: it depends on the required speed, the allowable stress on the mosfet, EMI, and the power involved.

Solution	Where energy is dissipated	Turn-off speed	Stress on the mosfet	EMI	When to use it
No protection	Breackdown mosfet	fast	very high	high	never
Flyback diode	coil resistance	slow	very low	very low	Relays, slow loads
Flyback + zener	Coil and zener	medium	low	medium	When faster turn-off is needed
Snubber on the mosfet	R snubber	medium	low	good	Fast circuits, PWM
Snubber on the coil	R snubber	medium	medio basso	good	EMI reduction